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SYSTEMS AND METHODS FOR HIGH ACCURACY FIXTURELESS ASSEMBLY

Abstract

An approach to positioning one or more robotic arms in an assembly system may be described herein. For example, a system for robotic assembly may include a first robot, a second robot, and a control unit. The control unit may be configured to receive a first target location proximal to a second target location. The locations may indicate where the robots are to position the features. The control unit may be configured to calculate a first calculated location of the first feature of the first subcomponent, measure a first measured location of the first feature of the first subcomponent, determine a first transformation matrix between the first calculated location and the first measured location, reposition the first feature of the first subcomponent to the first target location using the first robot, the repositioning based on the first transformation matrix.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/814,441, filed Jul. 22, 2022, which is a continuation of U.S. application Ser. No. 16/222,926, filed Dec. 17, 2018 and entitled “SYSTEMS AND METHODS FOR HIGH ACCURACY FIXTURELESS ASSEMBLY”, which applications are incorporated by reference herein in their entirety.

BACKGROUND

Field

[0002] The present disclosure relates to transport structures such as automobiles, trucks, trains, boats, aircraft, motorcycles, metro systems, and the like, and more specifically to techniques for performing operations with robotic arms.

Background

[0003] A transport structure such as an automobile, truck or aircraft employs a large number of interior and exterior nodes. These nodes provide structure to the automobile, truck, and aircraft, and respond appropriately to the many different types of forces that are generated or that result from various actions like accelerating and braking. These nodes also provide support. Nodes of varying sizes and geometries may be integrated into a transport structure, for example, to provide an interface between panels, extrusions, and/or other structures. Thus, nodes are an integral part of transport structures.

[0004] Most nodes must be coupled to, or interface securely with, another part or structure in secure, well-designed ways. In order to securely connect a node with another part or structure, the node may need to undergo one or more processes in order to prepare the node to connect with the other part or structure. For example, the node may be machined at an interface in order to connect with various other parts or structures. Further examples of processes include surface preparation operations, heat treatment, electrocoating, electroplating, anodization, chemical etching, cleaning, support removal, powder removal, and so forth.

[0005] In order to produce a transport structure (e.g., a vehicle, an aircraft, a metro system, etc.), one or more assembly operations may be performed after a node is manufactured. For example, a node may be connected with a part, e.g., in order to form a portion of a transport structure (e.g., a vehicle chassis, etc.). Such assembly may involve a degree of accuracy that is within one or more tolerance thresholds of an assembly system, e.g., in order to ensure that the node is securely connected with the part and, therefore, the transport structure may be satisfactorily produced.

[0006] When robotic apparatuses (e.g., robotic end-of-arm tool center point) perform assembly operations, the robotic apparatuses are to be accurately positioned in order for the assembly operations to be accurately performed. For example, a robotic arm with which a node is engaged may be positioned so that the node is accurately connected with a part. Thus, a need exists for an approach to correctly positioning at least one robotic apparatus (e.g., a robotic end-of-arm tool center point) with a degree of precision that is within tolerance threshold(s) of an assembly system

when performing various assembly operations.

SUMMARY

[0007] The present disclosure generally relates to assembly operations performed in association with the production of transport structures. Such assembly operations may include connection of nodes (e.g., additively manufactured nodes) with parts and/or other structures. Because transport structures are to be safe, reliable, and so forth, approaches to accurately performing various assembly operations associated with the production of transport structures may be beneficial. Such approaches to various assembly operations may be performed by at least one robotic arm that may be instructed via computer-generated instructions. Accordingly, a computer may implement various techniques to generate instructions for at least one robotic arm that causes the at least one robotic arm to be correctly positioned when performing various assembly operations.

[0008] In the present disclosure, systems and methods for positioning a robotic arm may be described. In one aspect, a method of robotic assembly includes receiving a first target location indicating where a first robot is to position a first feature of a first subcomponent. The first target location may be proximal to a second target location indicating where a second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location. The method of robotic assembly also includes calculating a first calculated location of the first feature of the first subcomponent and measuring a first measured location of the first feature of the first subcomponent. Additionally, the method of robotic assembly includes determining a first transformation matrix between the first calculated location and the first measured location and repositioning the first feature of the first subcomponent to the first target location using the first robot. The repositioning may be based on the first transformation matrix.

[0009] In one aspect, a system for robotic assembly includes a first robot, a second robot, and a control unit. The control unit may be configured to receive a first target location indicating where the first robot is to position a first feature of a first subcomponent. The first target location may be proximal to a second target location indicating where the second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location. The control unit may also be configured to calculate a first calculated location of the first feature of the first subcomponent and measure a first measured location of the first feature of the first subcomponent. Additionally, the control unit may be configured to determine a first transformation matrix between the first calculated location and the first measured location and reposition the first feature of the first subcomponent to the first target location using the first robot. The repositioning may be based on the first transformation matrix.

[0010] In one aspect, a robotic assembly control unit includes at least one processor and a memory coupled to the at least one processor. The memory includes instructions configuring the control unit to receive a first target location indicating where a first robot is to position a first feature of a first subcomponent. The first target location is proximal to a second target location indicating where a second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location. The memory also includes instructions configuring the control unit to calculate a first calculated location of the first feature of the first subcomponent and measure a first measured location of the first feature of the first subcomponent. Additionally, the memory includes instructions configuring the control unit to determine a first transformation matrix between the first calculated location and the first measured location and reposition the first feature of the first subcomponent to the first target location using the first robot. The repositioning

is based on the first transformation matrix.

[0011] In one aspect, a computer-readable medium stores computer executable code for robotic assembly. In an aspect, the computer-readable medium may be cloud-based computer-readable mediums, such as a hard drive on a server attached to the Internet. The code, when executed by a processor, causes the processor to receive a first target location indicating where a first robot is to position a first feature of a first subcomponent. The first target location may be proximal to a second target location indicating where a second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location. The code, when executed by a processor, causes the processor to calculate a first calculated location of the first feature of the first subcomponent and measure a first measured location of the first feature of the first subcomponent. The code, when executed by a processor, causes the processor to determine a first transformation matrix between the first calculated location and the first measured location and reposition the first feature of the first subcomponent to the first target location using the first robot. The repositioning is based on the first transformation matrix.

[0012] It will be understood that other aspects of mechanisms for realizing high accuracy fixtureless assembly with additively manufactured components and the manufacture thereof will become readily apparent to those skilled in the art from the following detailed description, wherein it is shown and described only several embodiments by way of illustration. As will be realized by those skilled in the art, the disclosed subject matter is capable of other and different embodiments, and its several details are capable of modification in various other respects, all without departing from the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a diagram illustrating an exemplary embodiment of certain aspects of a Direct Metal Deposition (DMD) 3-D printer.

[0014] FIG. 2 is a conceptual flow diagram of a 3-D printing process using a 3-D printer.

[0015] FIGS. 3A-D is a diagram illustrating exemplary powder bed fusion (PBF) systems during different stages of operation.

[0016] FIG. 4 is a diagram illustrating a perspective of a first assembly system including a plurality of robots acting as fixtures.

[0017] FIG. 5 is a diagram illustrating a perspective of a second assembly system including a plurality of robots acting as fixtures.

[0018] FIG. 6 is a diagram illustrating a fixture point printed directly on a part.

[0019] FIG. 7 is a diagram illustrating part scanning and fitting on a fixture.

[0020] FIG. 8 is a conceptual flow diagram in accordance with the systems and methods described herein.

DETAILED DESCRIPTION

[0021] The detailed description set forth below in connection with the appended drawings is intended to provide a description of various exemplary embodiments and is not intended to represent the only embodiments in which the invention may be practiced. The terms “exemplary,” “illustrative,” and the like used throughout the present disclosure mean “serving as an example, instance, or illustration,” and should not necessarily be construed as preferred or advantageous over other embodiments presented in the present disclosure. The detailed description includes specific details for the purpose of providing a thorough and complete disclosure that fully conveys the

scope of the invention to those skilled in the art. However, the invention may be practiced without these specific details. In some instances, well-known structures and components may be shown in block diagram form, or omitted entirely, in order to avoid obscuring the various concepts presented throughout the present disclosure. In addition, the figures may not be drawn to scale and instead may be drawn in a way that attempts to most effectively highlight various features relevant to the subject matter described.

[0022] Additive Manufacturing (3-D Printing). Additive manufacturing (AM) is advantageously a non-design specific manufacturing technique. AM provides the ability to create complex structures within a part. For example, nodes can be produced using AM. A node is a structural member that may include one or more interfaces used to connect to other spanning components such as tubes, extrusions, panels, other nodes, and the like. Using AM, a node may be constructed to include additional features and functions, depending on the objectives. For example, a node may be printed with one or more ports that enable the node to secure two parts by injecting an adhesive rather than welding multiple parts together, as is traditionally done in manufacturing complex products. Alternatively, some components may be connected using a brazing slurry, a thermoplastic, a thermoset, or another connection feature, any of which can be used interchangeably in place of an adhesive. Thus, while welding techniques may be suitable with respect to certain embodiments, additive manufacturing provides significant flexibility in enabling the use of alternative or additional connection techniques.

[0023] A variety of different AM techniques have been used to 3-D print components composed of various types of materials. Numerous available techniques exist, and more are being developed. For example, Directed Energy Deposition (DED) AM systems use directed energy sourced from laser or electron beams to melt metal. These systems utilize both powder and wire feeds. The wire feed systems advantageously have higher deposition rates than other prominent AM techniques. Single Pass Jetting (SPJ) combines two powder spreaders and a single print unit to spread metal powder and to print a structure in a single pass with apparently no wasted motion. As another illustration, electron beam additive manufacturing processes use an electron beam to deposit metal via wire feedstock or sintering on a powder bed in a vacuum chamber. Single Pass Jetting is another exemplary technology claimed by its developers to be much quicker than conventional laser-based systems. Atomic Diffusion Additive Manufacturing (ADAM) is still another recently developed technology in which components are printed, layer-by-layer, using a metal powder in a plastic binder. After printing, plastic binders are removed, and the entire part is sintered at once into a desired metal. One of several such AM techniques, as noted, is DMD. FIG. 1 illustrates an exemplary embodiment of certain aspects of a DMD 3-D printer **100**. DMD printer **100** uses feed nozzle **102** moving in a predefined direction **120** to propel powder streams **104a** and **104b** into a laser beam **106**, which is directed toward a workpiece **112** that may be supported by a substrate. Feed nozzle may also include mechanisms for streaming a shield gas **116** to protect the welded area from oxygen, water vapor, or other components.

[0024] The powdered metal is then fused by the laser **106** in a melt pool region **108**, which may then bond to the workpiece **112** as a region of deposited material **110**. The dilution area **114** may include a region of the workpiece where the deposited powder is integrated with the local material of the workpiece. The feed nozzle **102** may be supported by a computer numerical controlled (CNC) robot or a gantry, or another computer-controlled mechanism. The feed nozzle **102** may be moved under computer control multiple times along a predetermined direction of the substrate until an initial layer of the deposited material **110** is formed over a desired area of the workpiece **112**. The feed nozzle **102** can then scan the region immediately above the prior layer to deposit successive layers until the desired structure is formed. In general, the feed nozzle **102** may be configured to move with respect to all three axes, and in some instances to rotate on its own axis by a predetermined amount.

[0025] FIG. 2 is a flow diagram **200** illustrating an exemplary process of 3-D printing. A data

model of the desired 3-D object to be printed is rendered (operation **210**). A data model is a virtual design of the 3-D object. Thus, the data model may reflect the geometrical and structural features of the 3-D object, as well as its material composition. The data model may be created using a variety of methods, including CAE-based optimization, 3D modeling, photogrammetry software, and camera imaging. CAE-based optimization may include, for example, cloud-based optimization, fatigue analysis, linear or non-linear finite element analysis (FEA), and durability analysis.

[0026] 3-D modeling software, in turn, may include one of numerous commercially available 3-D modeling software applications. Data models may be rendered using a suitable computer-aided design (CAD) package, for example in an STL format. STL is one example of a file format associated with commercially available stereolithography-based CAD software. A CAD program may be used to create the data model of the 3-D object as an STL file. Thereupon, the STL file may undergo a process whereby errors in the file are identified and resolved.

[0027] Following error resolution, the data model can be “sliced” by a software application known as a slicer to thereby produce a set of instructions for 3-D printing the object, with the instructions being compatible and associated with the particular 3-D printing technology to be utilized (operation **220**). Numerous slicer programs are commercially available. Generally, the slicer program converts the data model into a series of individual layers representing thin slices (e.g., 100 microns thick) of the object being printed, along with a file containing the printer-specific instructions for 3-D printing these successive individual layers to produce an actual 3-D printed representation of the data model.

[0028] The layers associated with 3-D printers and related print instructions need not be planar or identical in thickness. For example, in some embodiments depending on factors like the technical sophistication of the 3-D printing equipment and the specific manufacturing objectives, etc., the layers in a 3-D printed structure may be non-planar and/or may vary in one or more instances with respect to their individual thicknesses.

[0029] A common type of file used for slicing data models into layers is a G-code file, which is a numerical control programming language that includes instructions for 3-D printing the object. The G-code file, or other file constituting the instructions, is uploaded to the 3-D printer (operation **230**). Because the file containing these instructions is typically configured to be operable with a specific 3-D printing process, it will be appreciated that many formats of the instruction file are possible depending on the 3-D printing technology used.

[0030] In addition to the printing instructions that dictate what and how an object is to be rendered, the appropriate physical materials necessary for use by the 3-D printer in rendering the object are loaded into the 3-D printer using any of several conventional and often printer-specific methods (operation **240**). In DMD techniques, for example, one or more metal powders may be selected for layering structures with such metals or metal alloys. In selective laser melting (SLM), selective laser sintering (SLS), and other PBF-based AM methods (see below), the materials may be loaded as powders into chambers that feed the powders to a build platform. Depending on the 3-D printer, other techniques for loading printing materials may be used.

[0031] The respective data slices of the 3-D object are then printed based on the provided instructions using the material(s) (operation **250**). In 3-D printers that use laser sintering, a laser scans a powder bed and melts the powder together where the structure is desired and avoids scanning areas where the sliced data indicates that nothing is to be printed. This process may be repeated thousands of times until the desired structure is formed, after which the printed part is removed from a fabricator. In fused deposition modeling, as described above, parts are printed by applying successive layers of model and support materials to a substrate. In general, any suitable 3-D printing technology may be employed for purposes of the present disclosure.

[0032] Another AM technique includes powder-bed fusion (“PBF”). Like DMD, PBF creates ‘build pieces’ layer-by-layer. Each layer or ‘slice’ is formed by depositing a layer of powder and exposing portions of the powder to an energy beam. The energy beam is applied to melt areas of the powder

layer that coincide with the cross-section of the build piece in the layer. The melted powder cools and fuses to form a slice of the build piece. The process can be repeated to form the next slice of the build piece, and so on. Each layer is deposited on top of the previous layer. The resulting structure is a build piece assembled slice-by-slice from the ground up.

[0033] FIGS. 3A-D illustrate respective side views of an exemplary PBF system **300** during different stages of operation. As noted above, the particular embodiment illustrated in FIGS. 3A-D is one of many suitable examples of a PBF system employing principles of the present disclosure. It should also be noted that elements of FIGS. 3A-D and the other figures in the present disclosure are not necessarily drawn to scale but may be drawn larger or smaller for the purpose of better illustration of concepts described herein. PBF system **300** can include a depositor **301** that can deposit each layer of metal powder, an energy beam source **303** that can generate an energy beam, a deflector **305** that can apply the energy beam to fuse the powder, and a build plate **307** that can support one or more build pieces, such as a build piece **309**. PBF system **300** can also include a build floor **311** positioned within a powder bed receptacle. The walls of the powder bed receptacle **312** generally define the boundaries of the powder bed receptacle, which is sandwiched between the walls **312** from the side and abuts a portion of the build floor **311** below. Build floor **311** can progressively lower build plate **307** so that depositor **301** can deposit a next layer. The entire mechanism may reside in a chamber **313** that can enclose the other components, thereby protecting the equipment, enabling atmospheric and temperature regulation and mitigating contamination risks. Depositor **301** can include a hopper **315** that contains a powder **317**, such as a metal powder, and a leveler **319** that can level the top of each layer of deposited powder.

[0034] Referring specifically to FIG. 3A, this figure shows PBF system **300** after a slice of build piece **309** has been fused, but before the next layer of powder has been deposited. In fact, FIG. 3A illustrates a time at which PBF system **300** has already deposited and fused slices in multiple layers, e.g., 150 layers, to form the current state of build piece **309**, e.g., formed of 150 slices. The multiple layers already deposited have created a powder bed **321**, which includes powder that was deposited but not fused.

[0035] FIG. 3B shows PBF system **300** at a stage in which build floor **311** can lower by a powder layer thickness **323**. The lowering of build floor **311** causes build piece **309** and powder bed **321** to drop by powder layer thickness **323**, so that the top of the build piece and powder bed are lower than the top of powder bed receptacle wall **312** by an amount equal to the powder layer thickness. In this way, for example, a space with a consistent thickness equal to powder layer thickness **323** can be created over the tops of build piece **309** and powder bed **321**.

[0036] FIG. 3C shows PBF system **300** at a stage in which depositor **301** is positioned to deposit powder **317** in a space created over the top surfaces of build piece **309** and powder bed **321** and bounded by powder bed receptacle walls **312**. In this example, depositor **301** progressively moves over the defined space while releasing powder **317** from hopper **315**. Leveler **319** can level the released powder to form a powder layer **325** that has a thickness substantially equal to the powder layer thickness **323** (see FIG. 3B). Thus, the powder in a PBF system can be supported by a powder support structure, which can include, for example, a build plate **307**, a build floor **311**, a build piece **309**, walls **312**, and the like. It should be noted that the illustrated thickness of powder layer **325** (i.e., powder layer thickness **323** (FIG. 3B)) is greater than an actual thickness used for the example involving **350** previously-deposited layers discussed above with reference to FIG. 3A.

[0037] FIG. 3D shows PBF system **300** at a stage in which, following the deposition of powder layer **325** (FIG. 3C), energy beam source **303** generates an energy beam **327** and deflector **305** applies the energy beam to fuse the next slice in build piece **309**. In various exemplary embodiments, energy beam source **303** can be an electron beam source, in which case energy beam **327** constitutes an electron beam. Deflector **305** can include deflection plates that can generate an electric field or a magnetic field that selectively deflects the electron beam to cause the electron beam to scan across areas designated to be fused. In various embodiments, energy beam source **303**

can be a laser, in which case energy beam **327** is a laser beam. Deflector **305** can include an optical system that uses reflection and/or refraction to manipulate the laser beam to scan selected areas to be fused.

[0038] In various embodiments, the deflector **305** can include one or more gimbals and actuators that can rotate and/or translate the energy beam source to position the energy beam. In various embodiments, energy beam source **303** and/or deflector **305** can modulate the energy beam, e.g., turn the energy beam on and off as the deflector scans so that the energy beam is applied only in the appropriate areas of the powder layer. For example, in various embodiments, the energy beam can be modulated by a digital signal processor (DSP).

[0039] The present disclosure presents various approaches to positioning at least one robotic arm in an assembly system. For example, an assembly system may include two robots, each of which may include a respective robotic arm. A first robotic arm may be configured to engage with a node during various operations performed with the node. For example, the first robotic arm may engage with a node that is to be connected with a part, and the part may be engaged by a second robotic arm. Various operations performed with the node (e.g., connecting the node with a part) may be performed with a relatively high degree of precision. Accordingly, at least one of the robotic arms may be positioned (e.g., repositioned) during an operation with the node in order to function in accordance with the precision commensurate with the operation.

[0040] In some aspects, the first robotic arm may engage with the node and the second robotic arm may engage with a part. An operation with the node may include connecting the node with the part. Thus, the first robotic arm may be positioned relative to the second robotic arm and/or the second robotic arm may be positioned relative to the first robotic arm. When the first and/or second robotic arms are configured to move, the first and/or second robotic arms may be positioned (e.g., repositioned) relative to the other one of the first and/or second robotic arms. Such positioning may correct the position(s) of the first and/or second robotic arms, e.g., to maintain the precision necessary for operations with a node, including connecting a node with a part by the first and second robotic arms.

[0041] The present disclosure provides various different embodiments of positioning one or more robotic arms of an assembly system for assembly processes and/or post-processing operations. It will be appreciated that various embodiments described herein may be practiced together. For example, an embodiment described with respect to one illustration of the present disclosure may be implemented in another embodiment described with respect to another illustration of the present disclosure.

[0042] FIG. **4** is a diagram illustrating a perspective of a first assembly system **400** including a plurality of robots **402**, **404** acting as fixtures for two nodes **406**, **408**. The assembly system **400** may be employed in various operations associated with the assembly of a node-based transport structure. In one embodiment, the assembly system **400** may perform at least a portion of the assembly of the node-based transport structure without any fixtures. For example, the assembly system **400** may be implemented for connecting a first node **406** (node 1) with a second node **408** (node 2) (although other implementations are possible without departing from the scope of the present disclosure). In an aspect, at least one of the first node **406** (e.g., first subcomponent) or the second node **408** (e.g., second subcomponent) may include a complex structure such as a chassis for a transport structure.

[0043] The assembly system **400** may include a first robotic arm **410** on the first robot **402** (robot 1). The first robotic arm **410** may have a distal end **414** and a proximal end **416**. The distal end **414** may be configured for movement, e.g., for operations associated with a node and/or part, e.g., the first node **406**. The proximal end **416** may secure the first robotic arm **410**, e.g., to a base **418**.

[0044] The distal end **414** of the first robotic arm **410** may be connected with a tool flange. The tool flange may be configured to connect with one or more components (e.g., tools) so that the first robotic arm **410** may connect with the one or more components and position the one or more

components as the first robotic arm **410** moves.

[0045] In the illustrated embodiment, the distal end **414** of the first robotic arm **410** may be connected with an end effector, e.g., by means of the tool flange. That is, the end effector may be connected with the tool flange, and the tool flange may be connected with the distal end **414** of the first robotic arm **410**. The end effector may be a component configured to interface with various parts, nodes, and/or other structures. Illustratively, the end effector may be configured to engage with a node **406** (however, the end effector may be configured to engage with a part or other structure). Examples of an end effector may include jaws, grippers, pins, or other similar components capable of engaging a node, part, or other structure.

[0046] As illustrated, the assembly system **400** may further include a second robotic arm **412** on the second robot **404**. The second robotic arm **412** may have a distal end **420** and a proximal end **422**. The proximal end **422** of the second robotic arm **412** may be connected with a base **424**, e.g., in order to secure the second robotic arm **412**. Illustratively, the first robotic arm **410** and the second robotic arm **412** may be located in the assembly system **400** to be approximately facing one another, e.g., so that the distal end **414** of the first robotic arm **410** extends towards the distal end **420** of the second robotic arm **412** and, correspondingly, the distal end **420** of the second robotic arm **412** extends towards the distal end **414** of the first robotic arm **410**. However, the first and second robotic arms **410**, **412** may be differently located in the assembly system **400** in other embodiments, e.g., according to an assembly operation that is to be performed.

[0047] Similar to the first robotic arm **410**, the distal end **420** of the second robotic arm **412** may be connected with a tool flange, and the tool flange may be connected with an end effector. The end effector may be configured to engage with a node, part, or other structure, such as the node **408** (node 2) that is to be connected with the node **406**.

[0048] Industrial robots may produce highly repeatable movements. For example, the robots **402**, **404** may be able to position each of the respective robot arms **410**, **412** repeatedly. The repeatability of the positioning may be accurate to about 60 microns when compared to another positioning, in an example. However, the placement relative to a specific positioning or absolute positioning may suffer from lower accuracy, e.g., approximately 400 microns.

[0049] Accordingly, because positioning of a robot arm **410**, **412** may be accurate relative to a previous positioning, but not as accurate relative to an absolute position, e.g., a specific x, y, z location where the robot may be directed by a control unit, robots **402**, **404** may generally not be suitable for use as high accuracy fixtures. The absolute positioning accuracy issue may be amplified when assembly hard points are driven by nominal location data. For example, parts tolerances may add in ways that negatively impact the tolerances of the assembled part when robotic positioning is used.

[0050] Metrology is the science of measurement. Using measurements to guide the robot, metrology guidance may be used to guide the robots **402**, **404** as assembly fixtures, as is discussed in greater detail with respect to FIG. 5. For example, metrology systems may offer accuracy in the range of approximately 30 microns. Using the guidance of a metrology system industrial robots can realize greatly improved accuracy (in the micron scale). With this improvement in accuracy, the tool center point (TCP) of industrial robots can be used as a high accuracy flexible fixture.

[0051] FIG. 5 is a diagram illustrating a perspective of a second assembly system **500** including a plurality of robots **502**, **504** acting as fixtures. The second assembly system **500** is generally similar to the assembly system **400** but includes further details of using measurements to guide the robot. Metrology guidance may be used to guide the robots **502**, **504** as assembly fixtures. The second assembly system **500** includes a metrology device **526**, metrology targets **528**, and cell frames **530**. The cell frames may define the work area and provide a frame of reference within the work area.

[0052] A metrology's system accuracy may be applied to a critical motion path segment of multiple robots. In an aspect, the fixtureless assembly process may include a cell reference frame that may be created using computer-aided design (CAD). The cell reference frame may be matched to a

physical robot cell.

[0053] The metrology device may be a metrology unit such as a laser, greyscale camera, or another device capable of taking measurements based on metrology targets. The metrology targets **528** may be mounted on robot flange and offset to the robot TCP.

[0054] Nominal target frames may be stored in the robot program, PLC, metrology software, or another database. Nominal frames may be dynamic and driven by scan results and/or probe results, as discussed with respect to FIG. **6** below. Each robot control unit **532** may be digitally connected to the metrology unit and metrology software.

[0055] In an aspect, the metrology process, in the context of assembling a node-based structure, may include a first robot sending a signal to a metrology unit to aim/focus on a critical location. The metrology unit may aim or focus on a location including a target and lock onto the target. For example, the metrology unit may use a small diameter scan and lock.

[0056] The metrology unit measures robot TCP location or another critical feature offset from a target. An aspect may compare a measured location value to a dynamic nominal location value. For example, a system may compare where a robot thinks a node is located to where the node is actually located. A system may then compute a transformation matrix to move from a current location to a goal location. The transformation matrix may be applied to a robot control unit/PLC **532**, and the robot may move to a desired location. A confirmation measurement may be performed. Accuracy boundaries may be adjustable, such as gain or other values minimize cycle time. Additionally, a second robot may send a signal to the metrology unit to aim/focus on a location. The process may continue and be repeated.

[0057] In an aspect, the control unit **532** may cause a scanner **534** to scan a first subcomponent to determine a relative location of the first feature of the first subcomponent relative to the TCP. A robot **502**, **504** may be configured to pick up the first subcomponent based on the scanning.

[0058] In an aspect, multiple metrology units and/or types of metrology units may be integrated to reduce cycle time. Measurements may also be taken in parallel with correction applied in parallel. Corrections may be applied only to the particular segments of the robot path. One metrology system may be used to apply a correction to n number of robots.

[0059] The control unit **532** illustrated in FIG. **5** may be a robotic assembly control unit **532**. The robotic assembly control unit **532** may include at least one processor and a memory coupled to the at least one processor. The memory may include instructions configuring the control unit **532** to receive a first target **528** location indicating where a first robot (e.g., robot 1) is to position a first feature of a first subcomponent. The first target **528** location proximal to a second target **528** location indicating where a second robot (e.g., robot 2) is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location.

[0060] The control unit **532** may calculate a first calculated location of the first feature of the first subcomponent and measure a first measured location of the first feature of the first subcomponent. Additionally, the control unit **532** may determine a first transformation matrix between the first calculated location and the first measured location and reposition the first feature of the first subcomponent to the first target location using the first robot, the repositioning based on the first transformation matrix. In an aspect, repositioning of the first feature of the first subcomponent and/or repositioning the second feature of the second subcomponent may be further based on a relative comparison of the first calculated location and the second calculated location. Accordingly, the features on the subcomponent may be located relative to each other directly, rather than relative to another reference frame rather than an absolute reference to a cell frame.

[0061] Accordingly, the control unit **532** may be coupled to the metrology device **526** and the robots (e.g., robot 1 and robot 2). Accordingly, measuring a first measured location of a first feature of a first subcomponent and measuring a second measured location of a second feature of a second

subcomponent may use the same metrology unit, e.g., the metrology device **526**. Using information from the metrology device **526**, the control unit **532** may calculate the first calculated location of the first feature of the first subcomponent and measure the first measured location of the first feature of the first subcomponent. The control unit **532** may then determine a first transformation matrix. The transformation matrix may be applied to adjust the position of a component held by the robot arm **510**, **512**. The measurements and calculations may be completed iteratively until the component or node is positioned as accurately as needed for the task. For example, two nodes may be located accurately enough to for a fixture to connect the two nodes.

[0062] The control unit **532** is illustrated in FIG. **5** as an individual unit coupled to the robots (robot 1 and robot 2) and the metrology device **526**. In other aspects, the control unit **532** may be made up of multiple sub-control units. These multiple sub-control units may be distributed among different devices. For example, the control unit **532** may be distributed between some combination of a separate unit, within one or more robots, and/or within one or more metrology devices **526**. For example, processing functionality may be located in the metrology device **526**, the first robot (robot 1), the second robot (robot 2), and an external control unit, e.g., coupled to the metrology device **526**, the first robot (robot 1), the second robot (robot 2).

[0063] FIG. **6** is a diagram illustrating a fixture point **602** printed directly on a part **600**. The part includes a part gripper portion **604**. An end of a robot arm **510**, **512** may grip the part at the part gripper portion **604**. Accordingly, the part **600** may be positioned by the robotic arm **510**, **512**. Thus, the control unit **532** may determine target **528** locations using measurements from the metrology device **526**. The control unit **532** may then control the robots (robot 1 and/or robot 2) to position components being held by the robot arms **510**, **512**.

[0064] The part **600** may be characterized, e.g., scanned, probed, or otherwise measured. As part of the characterization, features such as joints, bolt locations, or other features may be a fit to nominal data using a CAD model. For example, as part of the characterization, features such as joints, bolt locations, or other features may be a best fit to nominal data using a CAD model or other fit, e.g., any fit to determining an accurate characterization between two features. The characterization may measure the part relative to a TCP frame **606**. A best fit may be calculated based on the geometries of the features relative to the TCP frame **606**. Once the best fit of the features is performed, a fixture point **602** on the part may be calculated as a product of the calculated best fit. The fixture point **602** may allow the physical part to determine the fixture location that will lead to the most accurate assembly. The fixture point **602** may be printed directly into the product with the robot TCP acting as the fixture and a scanning fixture may be constructed with the robotic interface on it. Using the calculated best fit, adaptive fixture positions may be relocated in real time. The relocation of the fixture point may be driven by product geometry to maximize product accuracy and minimize overall assembly tolerance.

[0065] For example, the part **600** may include a sphere **608**. The fixture point **602** may be at the center of the sphere **608**. However, the sphere **608** may be imperfect. Accordingly, the part may be characterized to select a best location for the fixture point. The fixture point **602** may be an offset relative to the TCP frame.

[0066] The control unit **532** of FIG. **5** may cause a scanner **534** to scan a first subcomponent (e.g., part **600**) to determine a relative location of the first feature (e.g., fixture point **602**) of the first subcomponent (e.g., part **600**) relative to the TCP. The robot **502**, **504** may be configured to pick up the first subcomponent based on the scanning. For example, the robot **502**, **504** may pick up the first subcomponent at or near the TCP based on the scanning. In an aspect, the robot **502**, **504** may pick up the first subcomponent at or near the part gripper along the TCP based on the scanning. Accordingly, the system may be able to locate the first feature (e.g., fixture point **602**) of the first subcomponent (e.g., part **600**) relative to the TCP based on where the robot **502**, **504** picking up the first subcomponent (e.g., part **600**) based on the scanning.

[0067] FIG. **7** is a diagram illustrating part **700** scanning and fitting on a fixture. In the diagram

illustrates scanning the part **700** (1), determining a fit for the part **700** (2), and calculating a delta from a frame for the part **700**.

[0068] In an example, a fixture may have a feature which can easily be probed or scanned (1) to represent the robot TCP. For example, a sphere **708** with two flats **710** aligned so that its center is concentric with the center of the robot gripper/TCP **712**. The CAD file of each part includes the fixture **714** attached to each part **700**.

[0069] When determining the fit (2), the scan **750** may be of both the part **700** and the fixture **714**. The scan **750** is then overlaid with the CAD design **752** so that the features may be fit. The features may carry varying significance in the best fitting calculation. With just the main features, the fit may be used to determine a new location of the TCP **754**, e.g., at the center of the fixture sphere. The new location of the TCP **754** may be recorded as a calculated delta from a frame representing a CAD representation **756** of a part and a frame **758** based on an actual physical part with a newly located TCP **754**. The new TCP location **754** may be communicated in real time via a digital signal to an assembly cell software. The new location of the TCP **754** becomes the goal frame **758** in reference to a cell working frame **756** for the metrology system to correct the robot TCP to **754**. The goal frame **758** may be used in place of an ideal fixture location (TCP) from an idealized CAD design. The goal frame **758** may be based on product geometry and may be applied in real time to the assembly process of an actual physical part. For example, a best fit or other fit, e.g., a fit to determining an accurate characterization between two features.

[0070] FIG. **8** is a conceptual flow diagram in accordance with the systems and methods described herein. At **802** a control unit **532** may receive a first target location indicating where a first robot is to position a first feature of a first subcomponent. The first target location may be proximal to a second target location indicating where a second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the first feature of the first subcomponent in the first location and the second feature of the second subcomponent in the second location. The first target location may be a tool center point (TCP) and/or an offset from a TCP.

[0071] Accordingly, the control unit **532** receive target **528** location information from the metrology device **526**. The location information may indicate locations for nodes held by robot arms. For example, the location of the nodes may be known relative to the targets **528**. In an aspect, at least one of the first subcomponent may be a complex structure. The complex structure may be an automobile chassis.

[0072] At **804**, the control unit **532** may calculate a first calculated location of the first feature of the first subcomponent. The first calculated location may include a dynamic nominal location indicating a calculated location of a moving first feature at a specific time. The specific time may coincide with the measuring of the first location of the first feature.

[0073] For example, the control unit **532** may calculate a location using the location information received from the metrology device **526**. Because the location of the nodes may be known relative to the targets **528**, the control unit **532** may calculate a first calculated location of the first feature of the first subcomponent.

[0074] At **806**, the control unit **532** may measure a first measured location of the first feature of the first subcomponent. Measuring a first measured location of the first feature of the first subcomponent may include scanning the shape of the part. Additionally, scanning the shape of the part may include scanning the part (c.g., a first subcomponent) to determine a relative location of a first feature of the part relative to the TCP of the part. The first robot may be configured to pick up the part based on the scanning. For example, the first robot may pick up a first subcomponent on a TCP. Accordingly, the first robot may position the first feature based on the first feature's relative position to the TCP.

[0075] The first robot may signal a control unit **532** causing the control unit **532** to measure the first measured location of the first feature of the first subcomponent. In an aspect, measuring the

first measured location of the first feature of the first subcomponent and measuring the second measured location of the second feature of the second subcomponent use a same metrology unit. [0076] At **808**, the control unit **532** may determine a first transformation matrix between the first calculated location and the first measured location. Measuring a first measured location of the first feature of the first subcomponent comprises measuring a fixture point printed on the first subcomponent. For example, the control unit **532** may calculate frame deltas from an idealized frame based on a CAD design and a frame based on an actual physical device that may have differences from an idealized CAD design. (By idealized CAD design, the Applicant means the model design without inclusions of tolerances. The actual CAD design will generally include tolerances. A part that is made within tolerance may then be modeled as described herein using the frame delta. Parts not made within tolerances might be discarded.)

[0077] At **810**, the control unit **532** may reposition the first feature of the first subcomponent to the first target location using the first robot, the repositioning based on the first transformation matrix. Repositioning the first feature of the first subcomponent to the first target location using the first robot based on the first transformation matrix comprises sending the first transformation matrix to a control unit **532** in the first robot. Repositioning the first feature of the first subcomponent may be further based on a relative comparison of the first calculated location and a second calculated location.

[0078] In an aspect, the control unit **532** may repeat the calculating **804**, measuring **806**, determining **808**, and repositioning **810** steps. In another aspect, the control unit **532** may repeat one or more of **802**, **804**, **806**, **808**, and **810**. In an aspect, the repeating of one or more of **802**, **804**, **806**, **808**, and **810** may be relative to a second target. For example, the control unit **532** may receive a second target location indicating where the second robot is to position the second feature of the second subcomponent. The control unit **532** may calculate a second calculated location of the second feature of the second subcomponent. The control unit **532** may also measure a second measured location of the second feature of the second subcomponent. Additionally, the control unit **532** may determine a second transformation matrix between the second calculated location and the second measured location. The control unit **532** may also reposition the second feature of the second subcomponent to the second target location using the second robot. The repositioning may be based on the second transformation matrix.

[0079] At **812**, the control unit **532** may adjust at least one of accuracy boundaries or gain based on at least one of the repeating of the calculating, measuring, determining, and repositioning steps.

[0080] At **814**, the control unit **532** may characterize at least two features on the first subcomponent, the at least two features including the first target location.

[0081] At **816**, the control unit **532** may determine a fit, such as a best fit, for the at least two features. Repositioning the first feature of the first subcomponent to the first target location may use the first robot. The repositioning may be based on the first transformation matrix and may be further based on the best fit.

[0082] At **818**, the control unit **532** may attach the first subcomponent to the second subcomponent. Attaching the first subcomponent to the second subcomponent may include attaching the first subcomponent to the second subcomponent using an ultra-violet (UV) adhesive.

[0083] The present disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these exemplary embodiments presented throughout the present disclosure will be readily apparent to those skilled in the art, and the concepts disclosed herein may be applied to other techniques for printing nodes and interconnects. Thus, the claims are not intended to be limited to the exemplary embodiments presented throughout the disclosure but are to be accorded the full scope consistent with the language claims. All structural and functional equivalents to the elements of the exemplary embodiments described throughout the present disclosure that are known or later come to be known to those of ordinary skill in the art are intended to be encompassed by the claims. Moreover, nothing disclosed herein is

intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f), or analogous law in applicable jurisdictions, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

Claims

1.-42. (canceled)

43. A method, comprising: measuring a first feature of a first subcomponent; fitting the first feature to a corresponding feature of a data model of the first subcomponent; determining a fixture location of the first subcomponent based on the fitting; receiving a first target location indicating where a first robot is to position the fixture location of the first subcomponent, the first target location proximal to a second target location indicating where a second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the fixture location of the first subcomponent in the first target location and the second feature of the second subcomponent in the second target location; calculating a first calculated location of the fixture location; measuring a first measured location of the fixture location; determining a first transformation matrix between the first calculated location and the first measured location; and repositioning the fixture location to the first target location using the first robot, the repositioning based on the first transformation matrix.

44. The method of claim 43, wherein the first feature includes the fixture location.

45. The method of claim 43, wherein the first feature includes a joint.

46. The method of claim 43, wherein the first target location comprises a tool center point (TCP).

47. The method of claim 43, wherein measuring the first measured location comprises scanning the first subcomponent.

48. The method of claim 43, wherein measuring the first measured location comprises measuring a point on the first subcomponent.

49. The method of claim 43, further comprising repeating the calculating, the measuring of the measured location, the determining of the transformation matrix, and the repositioning.

50. The method of claim 43, further comprising adjusting at least accuracy boundaries or gain based on repeating of at least the calculating, the measuring of the measured location, the determining of the transformation matrix, or the repositioning.

51. The method of claim 43, wherein measuring the first measured location comprises a processor.

52. The method of claim 43, wherein the first calculated location comprises a dynamic nominal location indicating a calculated location of a moving fixture location at a specific time, the specific time coinciding with the measuring of a first location of the fixture location.

53. The method of claim 43, wherein repositioning the fixture location comprises sending the first transformation matrix to a control unit.

54. The method of claim 43, further comprising: measuring at least two locations on the first subcomponent, the at least two locations including the first target location, wherein the fitting includes determining a fit for the at least two locations, and wherein repositioning the fixture location is further based on the fit for the at least two locations.

55. The method of claim 43, wherein at least the first subcomponent or the second subcomponent comprises a plurality of components.

56. The method of claim 55, wherein the plurality of components comprises a chassis for a transport structure.

57. The method of claim 43, wherein the first target location comprises a location offset from a tool center point (TCP).

58. The method of claim 57, wherein measuring the first feature includes scanning the first subcomponent to determine a relative location of the fixture location of the first subcomponent

relative to the TCP, the first robot configured to pick up the first subcomponent based on the scanning.

59. The method of claim 43, further comprising attaching the first subcomponent to the second subcomponent.

60. The method of claim 59, wherein attaching the first subcomponent to the second subcomponent includes attaching the first subcomponent to the second subcomponent with an adhesive.

61. The method of claim 60, wherein the adhesive is an ultra-violet (UV) adhesive.

62. The method of claim 43, wherein the first feature includes a geometry of the first subcomponent.

63. The method of claim 62, wherein the geometry includes at least a surface of the first subcomponent.

64. The method of claim 43, wherein determining the fixture location includes performing a calculation of an offset relative to a tool center point (TCP) frame.

65. The method of claim 64, wherein performing the calculation is based on a best fit calculation.

66. The method of claim 64, wherein the TCP frame is defined on the first subcomponent.

67. The method of claim 43, wherein fitting the first feature includes performing a best fit to data.

68. The method of claim 67, wherein performing the best fit includes using the data model.

69. The method of claim 68, wherein the data model comprises a computer-aided design (CAD) model.

70. The method of claim 67, wherein performing the best fit includes performing a calculation based on a first geometry of the first feature relative to a tool center point (TCP) frame.

71. The method of claim 70, wherein the first feature includes a second feature, and wherein performing the calculation is based on a second geometry of the second feature relative to the TCP frame.

72. The method of claim 43, further comprising: receiving a second target location indicating where the second robot is to position the second feature of the second subcomponent; calculating a second calculated location of the second feature; measuring a second measured location of the second feature; determining a second transformation matrix between the second calculated location and the second measured location; and repositioning the second feature to the second target location using the second robot, repositioning of the second feature is based on the second transformation matrix.

73. The method of claim 72, wherein measuring the first measured location and measuring the second measured location use a same metrology unit.

74. The method of claim 72, wherein at least repositioning the fixture location or repositioning the second feature is based on a relative comparison of the first calculated location and the second calculated location.

75. A system, comprising: a first subcomponent comprising a first feature; a first robot; a second robot; and a control unit coupled to the first robot and the second robot, wherein the control unit includes a processor configured to: measure the first feature, fit the first feature to a corresponding feature of a data model of the first subcomponent, determine a fixture location of the first subcomponent based on the fit of the first feature to the corresponding feature, receive a first target location indicating where the first robot is to position the fixture location of the first subcomponent, the first target location proximal to a second target location indicating where the second robot is to position a second feature of a second subcomponent such that the first subcomponent and the second subcomponent form a component when coupled together with the fixture location of the first subcomponent in the first target location and the second feature of the second subcomponent in the second target location, calculate a first calculated location of the fixture location, measure a first measured location of the fixture location, determine a first transformation matrix between the first calculated location and the first measured location, and reposition the fixture location to the first target location using the first robot, the repositioning based on the first transformation matrix.

76. The system of claim 75, wherein the first feature includes the fixture location.
77. The system of claim 75, wherein the first feature includes a joint.
78. The system of claim 75, wherein the first target location comprises a tool center point (TCP).
79. The system of claim 75, wherein measuring the first measured location comprises scanning the first subcomponent.
80. The system of claim 75, wherein measuring the first measured location comprises measuring a point on the first subcomponent.
81. The system of claim 75, the control unit further configured to repeat the calculating, the measuring of the measured location, the determining of the transformation matrix, and the repositioning.
82. The system of claim 75, the control unit further configured to adjust at least accuracy boundaries or gain based on repeating of at least the calculating, the measuring of the measured location, the determining of the transformation matrix, or the repositioning.
83. The system of claim 75, wherein the first robot signals the control unit causing the control unit to measure the first measured location of the fixture location.
84. The system of claim 75, wherein the first calculated location comprises a dynamic nominal location indicating a calculated location of a moving fixture location at a specific time, the specific time coinciding with the measuring of a first location of the fixture location.
85. The system of claim 75, wherein repositioning the fixture location comprises sending the first transformation matrix to the control unit.
86. The system of claim 75, the control unit further configured to: characterize at least two features on the first subcomponent, the at least two features including the first target location, wherein the fit includes determine a fit for the at least two features, and wherein reposition the fixture location is further based on the fit for the at least two features.
87. The system of claim 75, wherein the control unit comprises a distributed control unit located in at least the first robot or the second robot.
88. The system of claim 75, wherein at least the first subcomponent or the second subcomponent comprises a plurality of components.
89. The system of claim 88, wherein the plurality of components comprises a chassis for a transport structure.
90. The system of claim 75, wherein the first target location comprises a location offset from a tool center point (TCP).
91. The system of claim 90, wherein measuring the first feature includes scanning the first subcomponent to determine a relative location of the fixture location of the first subcomponent relative to the TCP, the first robot configured to pick up the first subcomponent based on the scanning.
92. The system of claim 75, the control unit further configured to attach the first subcomponent to the second subcomponent.
93. The system of claim 92, wherein attaching the first subcomponent to the second subcomponent includes attaching the first subcomponent to the second subcomponent with an adhesive.
94. The system of claim 93, wherein the adhesive is an ultra-violet (UV) adhesive.
95. The system of claim 75, wherein the first feature includes a geometry of the first subcomponent.
96. The system of claim 95, wherein the geometry includes at least a surface of the first subcomponent.
97. The system of claim 75, wherein determining the fixture location includes a calculation of an offset relative to a tool center point (TCP) frame.
98. The system of claim 97, wherein the calculation of determining the fixture location is based on a best fit calculation.
99. The system of claim 97, wherein the TCP frame is defined on the first subcomponent.

- 100.** The system of claim 75, wherein the fit of the first feature includes a best fit to data.
- 101.** The system of claim 100, wherein the best fit includes using the data model.
- 102.** The system of claim 101, wherein the data model comprises a computer-aided design (CAD) model.
- 103.** The system of claim 100, wherein the best fit includes a calculation based on a first geometry of the first feature relative to a tool center point (TCP) frame.
- 104.** The system of claim 103, wherein the first feature includes a second feature, and wherein the calculation of the best fit is based on a second geometry of the second feature relative to the TCP frame.
- 105.** The system of claim 75, wherein the control unit further receives a second target location indicating where the second robot is to position the second feature of the second subcomponent; calculates a second calculated location of the second feature; measures a second measured location of the second feature; determines a second transformation matrix between the second calculated location and the second measured location; and repositions the second feature to the second target location using the second robot, repositioning of the second feature is based on the second transformation matrix.
- 106.** The system of claim 105, wherein measuring the first measured location and measuring the second measured location use a same metrology unit.
- 107.** The system of claim 105, wherein at least repositioning the fixture location or repositioning the second feature is based on a relative comparison of the first calculated location and the second calculated location.
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